

NATIONAL FORENSIC SCIENCES UNIVERSITY
PRACTICAL EXAMINATION MAY 2026

Date: 05/05/2026

Subject Code: CTMSDFIS SII L1

Total Marks: 50

Subject Name: Network Security and Forensics Lab / NSF LAB CTMSDFIS SII L1

Time: 30 Minutes

Instructions:

- Attempt all questions.
- Figures to the right indicate full marks.
- Identify your SET from the roll number allocation below.
- Assume suitable data wherever necessary.

SET-1: Roll Number ending with 01, 05, 09, 13, 17, 21, 25 **SET-2:** Roll Number ending with 02, 06, 10, 14, 18, 22, 26

SET-3: Roll Number ending with 03, 07, 11, 15, 19, 23, 27 **SET-4:** Roll Number ending with 04, 08, 12, 16, 20, 24

SET	Q.No.	Questions	Marks
SET-1 01, 05, 09 13, 17, 21, 25	Q.1	Capture network traffic generated while logging into https://www.saucedemo.com/ using browser developer tools or Wireshark. Identify request methods, response codes, and the protocol used. Explain whether sensitive information is exposed during transmission. Sample Data — Username: standard_user Password: secret_sauce	25
	Q.2	In the event of a security incident, such as a data breach or malware infection , describe the steps you would take to contain the incident, investigate the root cause, and communicate effectively with stakeholders.	
SET-2 02, 06, 10 14, 18, 22, 26	Q.1	Perform a vulnerability assessment on a https://ginandjuice.shop/ website to identify open ports, running services, and potential weaknesses. Compile a report with identified vulnerabilities and remediation suggestions. Tools: Nmap, Nessus, OpenVAS, Metasploit, etc. Objectives: Discover network vulnerabilities, evaluate risk levels, and propose security improvements to harden the network. Sample Data — Username carlos Password: hunter2	25
	Q.2	Analyze HTTPS communication for https://ginandjuice.shop/ . Inspect TLS version, cipher suite, and certificate details using browser security tools. Explain how weak TLS configurations could expose applications to downgrade or man-in-the-middle attacks. Sample Data — Username carlos Password: hunter2	
SET-3 03, 07, 11 15, 19, 23, 27	Q.1	Write a Python program that implements the RSA encryption algorithm . Demonstrate its usage by encrypting, decrypting and digital signature a given message.	25
	Q.2	Perform a basic reconnaissance scan on a given test system or localhost using Nmap . Identify open ports and running services. Classify discovered services into low, medium, or high risk, with justification.	
SET-4 04, 08, 12 16, 20, 24	Q.1	Given the following password hashes, attempt to identify weak passwords using appropriate tools or reasoning. Sample Data (MD5 hashes): 21232f297a57a5a743894a0e4a801fc3 5f4dcc3b5aa765d61d8327deb882cf99	25
	Q.2	Imagine a scenario where a distributed denial of service (DDoS) attack is targeting your organization's web servers. Outline the steps you would take to mitigate the impact and prevent future occurrences.	